Chapter 2.2 — Formula Sheet

Concentration Inequalities & Tail Bounds

Companion quick-reference to your Notebook 2 syllabus and your *20 Worked Examples* set (Example labels 1–20 cross-referenced below)

Classical Probability Bounds

Markov's Inequality [Ex 1–5]

For $X \geq 0$ and any $a > 0$:

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

Requires: non-negativity only — no variance, no shape.
General form (apply to a transform): for any increasing, non-negative ϕ,

$$\mathbb{P}(X \geq a) = \mathbb{P}(\phi(X) \geq \phi(a)) \leq \frac{\mathbb{E}[\phi(X)]}{\phi(a)}$$

e.g. $\phi(x) = x^2$ (sharper with known $\mathbb{E}[X^2]$), or $\phi(x) = e^{tx}$ (Chernoff bound):

$$\mathbb{P}(X \geq a) \leq \min_{t>0} e^{-ta}\, \mathbb{E}[e^{tX}]$$

Tightness: exact iff X is two-point, 0 or a.

Chebyshev's Inequality [Ex 6–10]

For any X with finite variance, any $k > 0$:

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Requires: finite variance only — works for negative-valued X (P&L, returns) where Markov cannot be applied directly.
Sample mean form: $\mathrm{Var}(\bar{X}_n) = \sigma^2/n \Rightarrow$

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2}$$

One-sided (Cantelli):

$$\mathbb{P}(X - \mu \geq k\sigma) \leq \frac{1}{1 + k^2}$$

(never just halve the two-sided bound — that assumes symmetry Chebyshev doesn't have).
Tightness: exact for a 3-point distribution at $\mu, \mu \pm k\sigma$ with masses $1 - \frac{1}{k^2}, \frac{1}{2k^2}, \frac{1}{2k^2}$.

Jensen's Inequality [Ex 11–14]

For convex g:

$$\mathbb{E}[g(X)] \geq g(\mathbb{E}[X])$$

For concave g, the inequality reverses: $\mathbb{E}[g(X)] \leq g(\mathbb{E}[X])$.
Key instances:

$$\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2 \;\Rightarrow\; \mathrm{Var}(X) \geq 0$$

$$\mathbb{E}[1/X] \geq 1/\mathbb{E}[X] \quad (X > 0)$$

$$\mathbb{E}[\max(S - K, 0)] \geq \max(\mathbb{E}[S] - K, 0)$$

Jensen's gap (2nd-order Taylor):

$$\mathbb{E}[g(X)] - g(\mu) \approx \frac{1}{2} g''(\mu)\, \mathrm{Var}(X)$$

Sign of g'' alone gives the gap's direction — convex $\Rightarrow \geq 0$, concave $\Rightarrow \leq 0$. Equality in Jensen iff X is a.s. constant or g is linear.

Limit Theorems

Weak Law of Large Numbers [Ex 15–17]

For i.i.d. $X_1, \ldots, X_n$ with mean μ, variance σ^2, and any fixed $\varepsilon > 0$:

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \longrightarrow 0 \quad \text{as } n \to \infty$$

Proof (via Chebyshev):

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \xrightarrow{n\to\infty} 0$$

What it does NOT say: convergence is of the *ratio* $\bar{X}_n$, not a "correction" of the raw count — $\mathbb{E}[|S_n - n\mu|]$ actually *grows* like $\Theta(\sqrt{n})$. Confusing this is the Gambler's Fallacy.

Central Limit Theorem [Ex 18–20]

For i.i.d. $X_1, \ldots, X_n$ with mean μ, variance σ^2:

$$S_n = \sum_{i=1}^{n} X_i \;\approx\; \mathcal{N}(n\mu,\; n\sigma^2)$$

$$\bar{X}_n \;\approx\; \mathcal{N}\left(\mu,\; \frac{\sigma^2}{n}\right), \qquad Z = \frac{S_n - n\mu}{\sigma\sqrt{n}} \;\approx\; \mathcal{N}(0, 1)$$

Continuity correction (discrete → continuous, e.g. Binomial sums):

$$\mathbb{P}(S_n \geq k) \approx 1 - \Phi\left(\frac{k - 0.5 - n\mu}{\sigma\sqrt{n}}\right)$$

Rate of convergence (Berry–Esseen):

$$\sup_x |F_n(x) - \Phi(x)| \leq \frac{C\rho}{\sigma^3\sqrt{n}}, \qquad \rho = \mathbb{E}[|X - \mu|^3]$$

Large skew/heavy tails (ρ large relative to σ^3) $\Rightarrow$ need much larger n before the Normal approximation is trustworthy — the "$n \geq 30$" rule of thumb silently assumes roughly symmetric, thin-tailed inputs.

One-tool-tree reminder: Markov needs only $X \geq 0$ and $\mathbb{E}[X]$; Chebyshev is Markov applied to $(X - \mu)^2$; WLLN is Chebyshev applied to $\bar{X}_n$ as $n \to \infty$; CLT sharpens WLLN's crude probability bound into an exact limiting shape. Four inequalities, one escalating idea: the more moments/structure you're willing to assume, the tighter the bound.